

Simon Kaplinsky

skaplins@andrew.cmu.edu | 610-864-0931 | linkedin.com/in/simon-kaplinsky

EDUCATION

Carnegie Mellon University

B.S. Electrical and Computer Engineering

Pittsburgh, PA

Expected May 2029

GPA: 3.84 / 4.00

Coursework completed ahead of standard sequence: Electronic Devices and Analog Circuits (18-220), Structure and Design of Digital Systems (18-240), Fundamentals of Electromagnetics (18-300), Signals and Systems (18-290), PCB Fabrication (18-021)

EXPERIENCE

Electrical Team Member

Sep 2025 – Present

CMU Rocket Command

Pittsburgh, PA

- Design, lay out, and debug custom circuits in KiCad under tight space and weight constraints, maintaining reliability through launch, flight, and landing loads
- Build an ESP32-based payload computer for NASA's University Student Launch Initiative, integrating an altimeter, conductivity sensor, motor driver, and SD-card data logging
- Write payload firmware that autonomously detects deployment and landing, then drives a motor to collect and measure a soil sample
- Perform board-level bring-up and fault isolation across repeated ground and flight tests

Teaching Assistant — Electronic Devices and Analog Circuits (18-220)

Aug 2026 – Present

Carnegie Mellon University, Department of ECE

Pittsburgh, PA

- Coach students through lab and homework debugging, guiding fault isolation in analog circuits and reasoning from measurement back to device behavior
- Lead small-group review sessions on core course topics and host weekly office hours

Software Engineering Summer Associate

May 2026 – Aug 2026

Byline Travel, Inc.

Palo Alto, CA

- Developed production Python and TypeScript for an AI-powered travel platform and optimized PostgreSQL-backed services, merging 40 pull requests over a 14-week internship
- Engineered an agentic web scraper to autonomously gather and structure point-of-interest data across 15 sources

PROJECTS

Interactive Wearable Costume

Jan 2026 – Present

Personal Project

- Design a sensor-to-actuator signal chain mapping heart rate, gyroscope, and capacitive touch input to addressable LED drivers in real time; hand-wire and solder all circuitry and write the device firmware

SKILLS

Circuits & PCB: Schematic capture, PCB layout and fabrication, analog/digital circuit analysis, embedded systems, sensor interfacing, board bring-up

Digital Design: RTL design, ALU and datapath design, FSMs, CPU architecture, CMOS modeling

Tools: KiCad, ESP32, Arduino, oscilloscope, logic analyzer, Git/GitHub, SolidWorks, PostgreSQL

Languages: C, C++, Python, TypeScript

ACTIVITIES & LEADERSHIP

Social Chair, Mr. Yuk Ultimate Frisbee

May 2026 – Present

Carnegie Mellon University

- Recruited 80+ prospective players through campus outreach to fill a 24-person competitive roster; manage team outreach and run official university events during orientation week